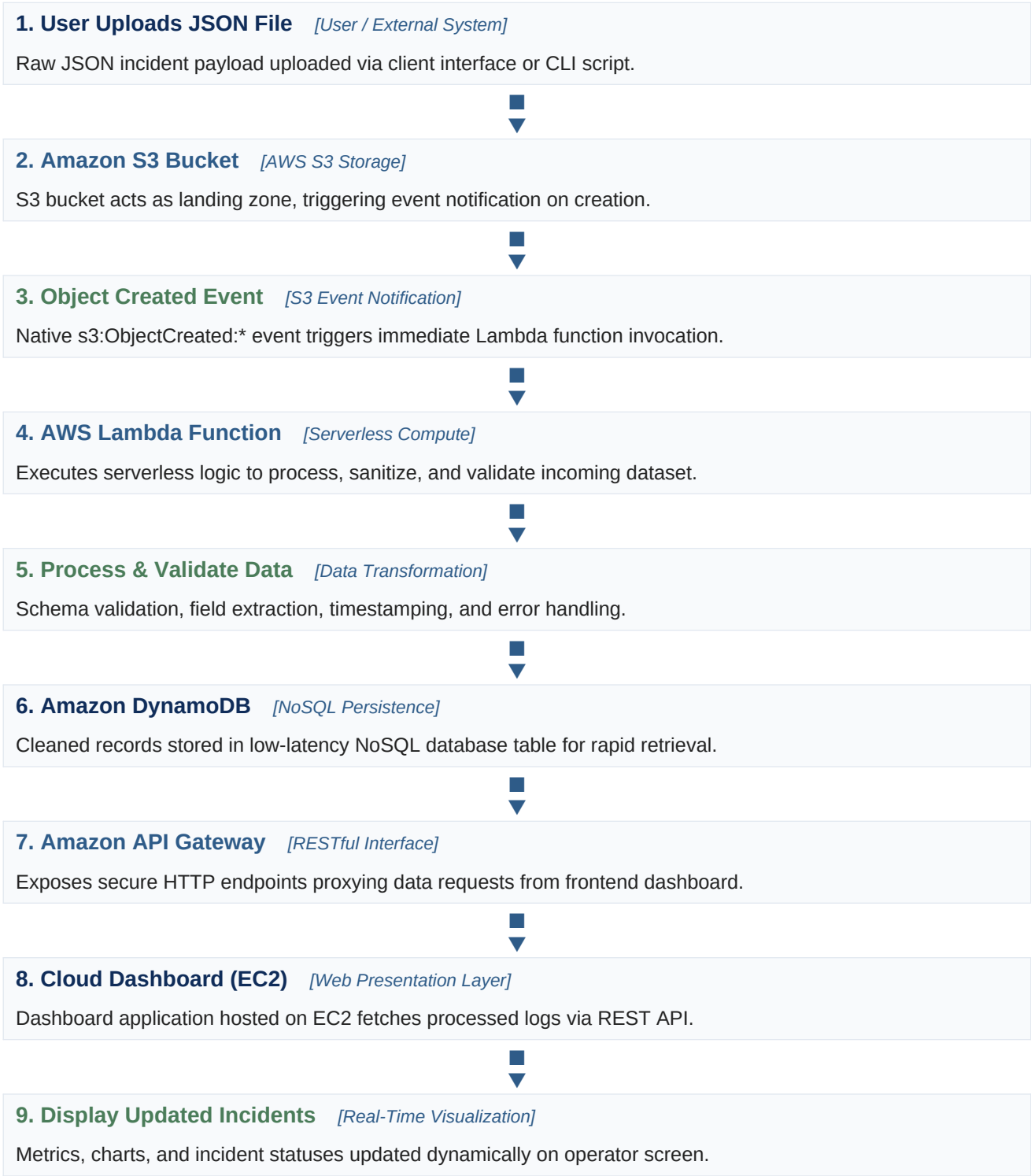


System Workflow Diagram

Cloud Incident Monitoring Dashboard — Event-Driven Data Pipeline

1. Architectural Data Flow Diagram

The visual flowchart below details the sequential processing stages from file ingestion to web dashboard update:



2. Workflow Stage Breakdown Matrix

Stage Name	Service Layer	Operational Purpose
1. User Uploads JSON File	User / External System	Raw JSON incident payload uploaded via client interface or CLI script.
2. Amazon S3 Bucket	AWS S3 Storage	S3 bucket acts as landing zone, triggering event notification on creation.
3. Object Created Event	S3 Event Notification	Native s3:ObjectCreated:* event triggers immediate Lambda function invocation.
4. AWS Lambda Function	Serverless Compute	Executes serverless logic to process, sanitize, and validate incoming dataset.
5. Process & Validate Data	Data Transformation	Schema validation, field extraction, timestamping, and error handling.
6. Amazon DynamoDB	NoSQL Persistence	Cleaned records stored in low-latency NoSQL database table for rapid retrieval.
7. Amazon API Gateway	RESTful Interface	Exposes secure HTTP endpoints proxying data requests from frontend dashboard.
8. Cloud Dashboard (EC2)	Web Presentation Layer	Dashboard application hosted on EC2 fetches processed logs via REST API.
9. Display Updated Incidents	Real-Time Visualization	Metrics, charts, and incident statuses updated dynamically on operator screen.